CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 7: **How do Organisms Reproduce?** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In which of the following does the offspring differ least from the parent? 1

(a) Offspring produced by seeds **(b)** Offspring produced by cross-pollination
(c) Offspring produced by budding in Hydra **(d)** Offspring produced by fertilisation

2. The thread-like structure on which anther is held is the 1

(a) filament **(b)** style **(c)** stigma **(d)** ovule

3. How many chromosomes would a human egg cell have, if the body cell has 46? 1

(a) 46 **(b)** 12 **(c)** 92 **(d)** 23

4. A unicellular organism that divides into many cells inside a protective cyst is 1

(a) Yeast **(b)** Paramecium **(c)** Plasmodium **(d)** Amoeba

5. A flower has only stamens. It is 1

(a) bisexual **(b)** unisexual (male) **(c)** sterile **(d)** female

6. The zygote in humans has 1

(a) 46 chromosomes **(b)** 12 chromosomes **(c)** 92 chromosomes **(d)** 23 chromosomes

7. Which statement about variation is correct? 1

(a) Only sexual reproduction ever has variation
(b) Variation harms every species
(c) Inaccurate DNA copying produces variation
(d) DNA copying is always perfect

8. An ovary has 20 ovules and all of them are fertilised. The fruit will have 1

(a) 2 seeds **(b)** 1 seed **(c)** 20 seeds **(d)** 0 seeds

9. The structure that connects the embryo to the placenta is the 1

(a) cervix **(b)** oviduct **(c)** urethra **(d)** umbilical cord

10. A potato tuber has several eyes. Each eye can grow into a plant by 1

(a) vegetative propagation **(b)** regeneration
(c) budding **(d)** spore formation

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do single-celled organisms reproduce by fission while many-celled complex organisms do not? 2

12. Why are variations not seen to a large extent in organisms reproducing asexually? 2

13. What is the role of the stigma being sticky and feathery in many flowers? 2

14. Why does a bread piece left in a warm moist place get covered with a woolly mass? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why a large number of spores are produced by Rhizopus even though only a few land in a suitable place. 3

16. Why is there a wide difference in the number of male and female gametes in many organisms? 3

17. What are the different ways in which asexual reproduction may help plants in agriculture? 3

OR

A farmer wants a banana plant that is identical to a good plant. Which method should he use and why? 3

18. Why are the sperms of mammals motile while eggs are not? 3

OR

What is the significance of variation in survival? Give an example. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe different methods of asexual reproduction in unicellular and multicellular organisms. State why they are not used by complex animals. 5

OR

Explain the structure and functions of the human female reproductive system, and how it supports the development of a baby. 5

20. A student places Bryophyllum leaves, a rose cutting and a potato eye in moist soil. Explain what happens in each and the type of reproduction involved. 5

OR

Explain what happens in the human female every month if 5
(a) the egg is fertilised and
(b) the egg is not fertilised.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND